Department of Computer Science

Course Name & No.: Cloud Computing 1501438

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Ch 5- Cloud Computing Services

Assignment 5

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Task 1: Compare SPI Service Models

Complete the table below by describing the level of control users have in each cloud service model (IaaS, PaaS, SaaS) and provide one example service for each.

Service Model	Level of Control	Example Service
IaaS		
PaaS		
SaaS		

Task 2: Identify the Service Model

For each scenario below, identify whether it describes IaaS, PaaS, or SaaS:

1. A developer builds an application using Google App Engine without managing servers.
   Model:

2. A company rents virtual machines and storage from AWS to host its own database and OS.
   Model:

3. An employee accesses email and documents through a web browser using Office 365.
   Model:

4. A startup uses Microsoft Azure to deploy a web app without worrying about underlying infrastructure.
   Model:

Task 3: Explain Virtualization in IaaS
What is virtualization and how does it enable Infrastructure-as-a-Service (IaaS)?
Provide two benefits of using virtualization in cloud computing.

Definition:
Benefit 1:
Benefit 2:

Task 4: Other Cloud Services (XaaS)
Match the following specialized cloud services with their descriptions:

Answer	Service	Description
	1. Storage-as-a-Service	A. Provides identity management, authentication, and access control.
	2. Database-as-a-Service	B. Offers data backup and recovery as a cloud service.
	3. Security Management-as-a-Service	C. Allows users to store files in the cloud like a virtual disk.
	4. Backup-as-a-Service	D. Provides managed database services without manual administration.

Task 5: Vendor Lock-in in PaaS
What is vendor lock-in in the context of Platform-as-a-Service (PaaS)?
Explain why it can be a problem and suggest one way to mitigate it.

Definition:
Problem:
Mitigation:

Task 1 :

Service Model	Level of Control	Example Service
IaaS (Infrastructure as a Service)	High control: user manages OS, storage, apps, and runtime; provider manages hardware	Amazon EC2
PaaS (Platform as a Service)	Medium control: user manages applications and data; provider manages OS, runtime, and infrastructure	Google App Engine
SaaS (Software as a Service)	Low control: user only uses the software; provider manages everything	Microsoft Office 365

Task 2:

 A developer builds an application using Google App Engine without managing servers.
Model: PaaS

 A company rents virtual machines and storage from AWS to host its own database and OS.
Model: IaaS

 An employee accesses email and documents through a web browser using Office 365.
Model: SaaS

 A startup uses Microsoft Azure to deploy a web app without worrying about underlying infrastructure.
Model: PaaS

Task 3:

Definition:
Virtualization is the process of creating virtual versions of physical resources (like servers, storage, and networks) using software, allowing multiple virtual machines to run on a single physical machine.

Benefit 1:
Efficient resource utilization – multiple users can share the same hardware, reducing cost.

Benefit 2:
Scalability and flexibility – resources can be quickly allocated or adjusted based on demand.

Task 4:

Answer	Service	Description
1	Storage-as-a-Service	**C**. Allows users to store files in the cloud like a virtual disk
2	Database-as-a-Service	**D**. Provides managed database services without manual administration
3	Security Management-as-a-Service	**A**. Provides identity management, authentication, and access control

Task 5:

Definition:
Vendor lock-in occurs when a customer becomes dependent on a specific cloud provider's platform, tools, or services, making it difficult to switch to another provider.

Problem:
It limits flexibility and can increase costs or technical challenges if the user wants to migrate to another platform.

Mitigation:
Use open standards and portable technologies (e.g., containers like Docker) to make applications easier to move between providers.